

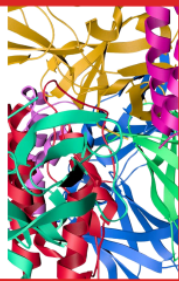
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CCR2-Mediated Uptake of Constitutively Produced CCL2: A Mechanism for Regulating Chemokine Levels in the Blood

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C-C chemokine receptor 2 (CCR2) is a key driver of monocyte/macrophage trafficking to sites of inflammation and has long been considered a target for intervention in autoimmune disease. However, systemic administration of CCR2 antagonists is associated with marked increases in CCL2, a CCR2 ligand, in the blood. This heretofore unexplained phenomenon complicates interpretation of in vivo responses to CCR2 antagonism. We report that CCL2 elevation after pharmacological CCR2 blockade is due to interruption in a balance between CCL2 secretion by a variety of cells and its uptake by constitutive internalization and recycling of CCR2. We observed this phenomenon in response to structurally diverse CCR2 antagonists in wild-type mice, and also found substantially higher CCL2 plasma levels in mice lacking the CCR2 gene. Our findings suggest that CCL2 is cleared from blood in a CCR2-dependent but G protein (G_{α_i} , G_{α_s} or $G_{\alpha_{q/11}}$)-independent manner. This constitutive internalization is rapid: on a given monocyte, the entire cell surface CCR2 population is turned over in <30 minutes. We also found that constitutive receptor internalization/recycling and ligand uptake are not universal across monocyte-expressed chemokine receptors. For example, CXCR4 does not internalize constitutively. In summary, we describe a mechanism that explains the numerous preclinical and clinical reports of increased CCL2 plasma levels following in vivo administration of CCR2 antagonists. These findings suggest that constitutive CCL2 secretion by monocytes and other cell types is counteracted by constant uptake and internalization by CCR2-expressing cells. The effectiveness of CCR2 antagonists in disease settings may be dependent upon this critical equilibrium. *The Journal of Immunology*, 2019, 203: 3157–3165.

The C-C chemokine receptor 2 (CCR2) is a G protein-coupled receptor that mediates the migration of leukocytes, most notably monocytes, into inflammatory sites (1). The interaction between CCR2 and its signature ligand, MCP-1 (also known as CCL2), has been thoroughly studied in humans and rodents and has long been considered an important clinical target for various chronic inflammatory disorders and as a novel approach for multiple forms of kidney disease (2–6). More recently, high levels of CCR2 were identified on subsets of monocytic myeloid-derived suppressor cells (7), which are major components of the tumor microenvironment that prevent cytotoxic T cells from killing tumor

cells. The possibility that CCR2 antagonists could prevent entry of monocytic myeloid-derived suppressor cells into tumors prompted clinical trials in pancreatic cancer, which yielded promising results (8, 9).

Studies evaluating CCR2 antagonists in both clinical and preclinical settings have revealed a consistent and unexplained phenomenon in which CCL2 becomes elevated in the blood of patients, primates, or rodents after treatment with CCR2 antagonists (3, 6, 10, 11). This increased concentration of CCL2 in the plasma could potentially counteract the effects of CCR2 blockade (3, 6), thus limiting the effectiveness of the drug. In the current study, we sought to understand the mechanism by which treatment with CCR2 antagonists results in increased levels of CCL2 in the blood. We used two structurally distinct CCR2 antagonists, MK-0812 (12, 13) and CCX598 (14), to fully evaluate their effects on plasma CCL2 levels and compared these findings to plasma levels from mice genetically deficient in CCR2. Further, we performed extensive in vitro experiments to identify the cellular sources of elevated CCL2 following CCR2-antagonist treatment and to determine how cells can continually remove extracellular CCL2 under basal conditions.

In this study, we report that human monocytes and other cells constitutively secrete CCL2, and that CCR2 is constitutively internalized and recycled, which removes CCL2 from the cellular environment. Conversely, CCL2 levels rise if CCL2 binding to CCR2 is blocked by an antagonist or if CCR2 is absent. The constitutive internalization and recycling of CCR2 thus provides an effective mechanism for regulating CCL2 levels in the blood or in an inflammatory microenvironment.

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The online version of this article contains supplemental material.

Abbreviations used in this article: ACKR, atypical chemokine receptor; CCR2, C-C chemokine receptor 2; $G_{q/11}$ KO, HEK 293 cells lacking functional $G_{q/11}$; h, human; KO, knockout; MFI, mean fluorescence intensity; PTX, pertussis toxin; siRNA, small interfering RNA; WT, wild-type.

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Materials and Methods

Isolation and culture of monocytes

PBMCs were isolated from leukocyte reduction system chambers from a Trima Accel blood collector. Blood from leukocyte reduction system chambers was diluted 1:4 (vol/vol) with calcium- and magnesium-free PBS, and PBMCs were enriched by Ficoll gradient centrifugation. Monocytes were isolated by CD14⁺ positive selection using a MACS system with human (h) CD14 MicroBeads (Miltenyi Biotec, Bergisch Gladbach, Germany), according to the manufacturer's protocol. Freshly isolated monocytes were plated into 48-well plates (Thermo Fisher Scientific, Roskilde, Denmark), and cultured in a 5% CO₂ incubator at 37°C for 24 h at a density of 1×10^6 cells/ml in RPMI 1640 containing 0.3 g/l L-glutamine (Cellgro; Mediatech, Herdon, VA) supplemented with 10% (v/v) FBS (Sigma-Aldrich), 10 mM HEPES (Cellgro; Mediatech, Tewksbury, MA) and 1 mM Sodium pyruvate (Cellgro; Mediatech).

Cell culture

HEK 293 cells lacking functional G α_s (Gs KO [knockout]) or G $\alpha_{q/11}$ (G $\alpha_{q/11}$ KO), prepared by CRISPR/Cas9, as previously reported (15, 16), and parental control HEK 293 wild-type (WT) cells, were a kind gift of Dr. Asuka Inoue (Tohoku University, Sendai, Japan). Cell lines were cultured in DMEM with GlutaMAX (Life Technologies) supplemented with 10% FBS and grown at 37°C with 5% CO₂. Stable CCR2-expressing cells were generated in the parental, G α_s , or G $\alpha_{q/11}$ KO HEK 293 lines by transfection of pReceiver-M02-CCR2b plasmid (GeneCopoeia Rockville, MD), followed by selection with G418 (Life Technologies Carlsbad, CA).

In vivo studies

Animals were purchased and housed in accordance with ChemoCentryx Institutional Animal Care and Use Committee guidelines and requirements. Female C57BL/6 mice were purchased from The Jackson Laboratory (Bar Harbor, ME). Female CCR2 KO mice (1) were bred and raised in the ChemoCentryx animal housing facility. C57BL/6 mice were divided into six groups ($n = 5$): MK-0812 or CCX598 were dosed by mouth at 0.1, 10, and 30 mg/kg for MK-0812, or at 0.5, 30, and 90 mg/kg for CCX598. One

group of CCR2 KO mice and one group of C57BL/6 were injected with only vehicle to serve as negative controls. All the mice were bled retro-orbitally 4 h after injection of the test agent. The mouse plasma was collected from EDTA blood.

In time course experiments, 10 mg/kg MK-0812 or 30 mg/kg CCX598 was administered to C57BL/6 mice ($n = 5$), and blood was collected as above for plasma at 2, 4, 6, and 24 h after dosing. Both CCL2 levels and drug concentration were measured. Drug plasma levels were analyzed by liquid chromatography/mass spectrometry at ChemoCentryx.

In vitro experiments: CCL2 accumulation in monocyte cultures in the presence of CCR2 antagonists

Freshly isolated monocytes were challenged with increasing doses of CCX598, MK-0812, CCX872, CCX140, INCB3344, or PF04634817 (only CCX598 and MK-0812 were discussed in the text, the remainder are described in Supplemental Fig. 1). After 24 h of incubation at 37°C with 5% CO₂ or at 4°C, treated and untreated monocyte supernatants were collected for evaluation of CCL2 levels. In the signal transduction study, monocytes were preincubated with 200 ng/ml pertussis toxin (PTX) for 2 h. After incubation, the cells were resuspended in culture medium and stimulated with the various CCR2 antagonists for 24 h at 37°C with 5% CO₂. In the cycloheximide study, the monocytes were treated with 50 μ g/ml cycloheximide for 2 h in culture medium and then stimulated with various concentrations of CCR2 antagonists for 24 h at 37°C with 5% CO₂. After 24 h of incubation, treated and untreated monocyte supernatants were collected for CCL2 determination. In the in vitro experiments in which chemokines were added exogenously, the freshly isolated monocytes were treated with CCX598 alone or in combination with the CCR1 inhibitor CCX721 (17) followed by addition of 1 nM CCL8, CCL13, or CCL7, then incubated together for 24 h at 37°C with 5% CO₂. Samples of supernatant were collected after 24 h of incubation and CCL8, CCL7, and CCL13 levels were determined by ELISA.

ELISA analysis

Chemokine levels in the supernatants of cultured cells were measured using commercially available ELISA kits (CCL2 DuoSet ELISA; R&D Systems,

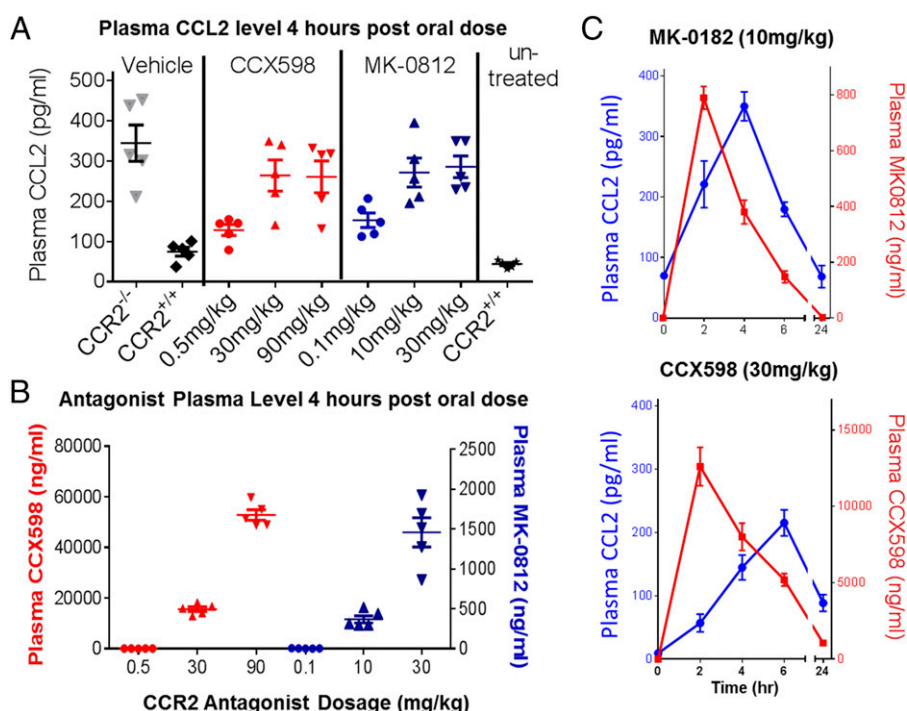


FIGURE 1. Antagonism of CCR2 and deletion of the gene encoding CCR2 both lead to elevated CCL2 levels in plasma. **(A)** C57BL/6 mice were dosed by oral gavage with the indicated doses of CCR2 antagonists CCX598 (red symbols) or MK-0812 (blue symbols). Mouse blood was collected from the retro-orbital plexus 4 h after dosing, and plasma CCL2 levels were determined by ELISA. WT C57BL/6 control mice receiving only vehicle (1% hydroxypropyl methylcellulose, black diamonds) were directly compared with gender and age-matched CCL2-deficient mice on the C57BL/6 background (also vehicle-injected, gray triangles). Resting plasma CCL2 levels were also measured for untreated C57BL/6 mice (black stars). **(B)** Plasma levels of CCR2 antagonists CCX598 and MK-0812 from the same blood samples used to measure CCL2 concentrations in **(A)** as determined by liquid chromatography with tandem mass spectrometry. **(C)** Time course for CCL2 (blue lines) and CCR2 antagonist (red lines) plasma levels after oral dosing with CCX598 (bottom panel) or MK-0812 (top panel). Each data point in **(C)** represents mean of five mice $\pm$ SEM.

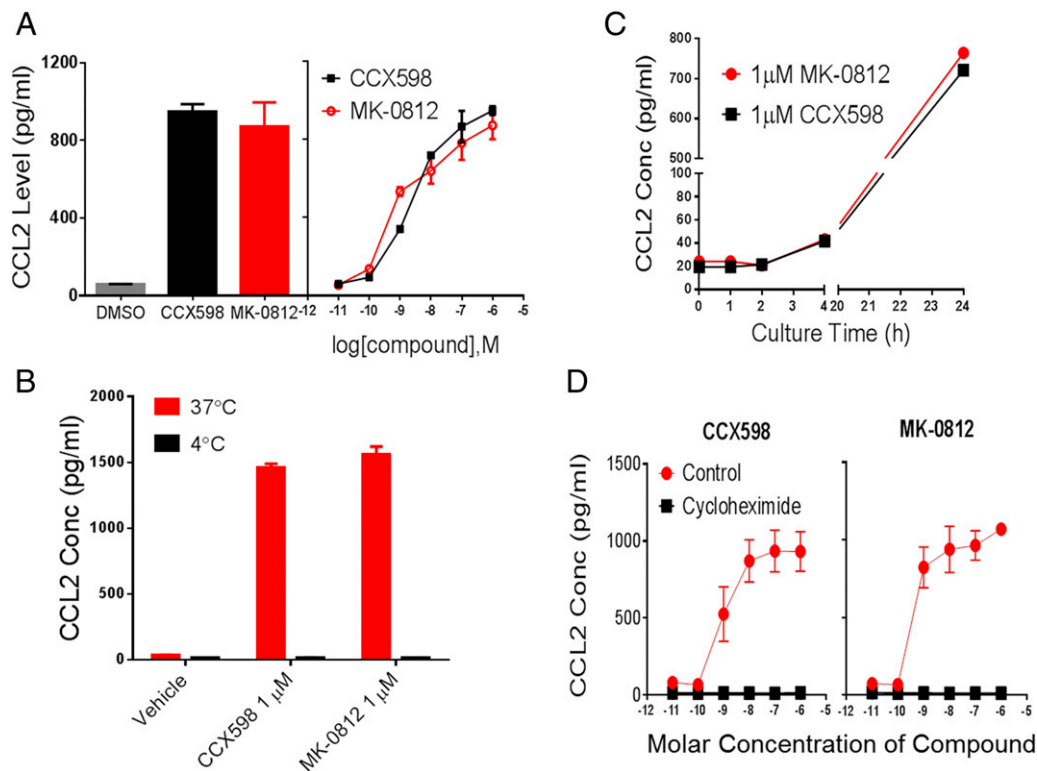


FIGURE 2. Accumulation of CCL2 following CCR2 antagonist treatment is time and temperature dependent and requires protein synthesis. **(A)** Left panel, Fresh human peripheral blood monocytes were incubated with vehicle (DMSO, gray bar), CCX598 (1 μ M) (black bar) or MK-0812 (1 μ M) (red bar) at 37°C for 24 h. Right panel, Fresh human peripheral blood monocytes were incubated with increasing concentrations of CCX598 or MK-0812 as indicated on the x-axis. The resulting CCL2 concentrations in the conditioned medium are indicated on the y-axis. **(B)** Monocytes were incubated at 37°C (red bars) or 4°C (black bars) with 1 μ M of CCR2 antagonists or vehicle. Data from a single experiment is shown, which used cells from a single donor representative of at least two other donors. **(C)** Monocytes were incubated with CCX598 (1 μ M) (black squares) or MK-0812 (1 μ M) (red circles) at 37°C for the incubation times indicated on the x-axis. CCL2 concentration in the conditioned medium is shown on the y-axis. **(D)** Monocytes were incubated as in (A)–(C) except for a 2 h preincubation with 50 μ g/ml cycloheximide (black squares) or vehicle (red circles). Cycloheximide-treated cells were shown to remain viable by demonstrating their ability to migrate in a gradient of CCL2 (see Supplemental Fig. 1).

Minneapolis, MN), (CCL8, CCL7, and CCL-13 Quantikine Human Kit; R&D Systems), following the manufacturer's instructions and were read with a FlexStation 3 plate reader (Molecular Devices, Sunnyvale, CA).

Chemotaxis assay

In Supplemental Fig. 2, freshly isolated monocytes were pretreated with 200 ng/ml PTX or 50 μ g/ml cycloheximide for 2 h. Chemotaxis assays were carried out by using ChemoTX chemotaxis chambers (Neuro Probe, Gaithersburg, MD). Cells were harvested by centrifugation and resuspended in chemotaxis buffer consisting of HBSS with 0.1% BSA at a density of 5×10^6 cells/ml. CCL2 was added to the bottom of the chambers and covered with a 5- μ m-pore-sized polycarbonate membrane filter, whereas the monocytes were added to the top of the filter. After 60 min incubation at 37°C, the assay was terminated by removal of cell drops from the top of the filter. Migration signal was determined by adding 5 μ l of CyQUANT solution (Invitrogen) to each well in the lower chemotaxis chamber and measuring the intensity of fluorescence on a SpectraFluor Plus plate reader (Tecan, Grödig, Austria).

Receptor internalization assay

CCR2R expression can be determined by flow cytometry, which measures the number of labeled CCR2 molecules remaining on the cell surface before ("prelabel") or after ("postlabel") 30 min of incubation at 37°C. In the prelabel experiments, monocytes were labeled with mouse anti-hCCR2 (clone number K036C2; BioLegend) or anti-hCXCR4 (clone number 12G5; R&D Systems) or its isotype-matched control for 30 min on ice and protected from light. Unbound Ab was washed away with wash buffer (HBSS, 1% FBS). Cells were further treated with 1 μ M MK-0812, 1 μ M CCX598, or vehicle control (DMSO) for 30 min at 37°C or pretreated with PTX or PBS as described above. After incubation, cells were transferred to wet ice, and the remaining surface receptor was labeled with anti-mouse Ab conjugated to allophycocyanin (715-136-151; Jackson ImmunoResearch). CCR2 or CXCR4 expression was analyzed on

a FACS LSRFortessa flow cytometer (BD Bioscience). Alternatively, in the postlabel internalization experiment setup, cells were labeled after 30 min of incubation at 37°C. To establish the time course for constitutive internalization, cells were labeled with CCR2 primary Ab prior to incubation at 37°C for 5, 10, 15, 30, 45, or 60 min. The internalization was stopped by transferring the tubes to wet ice. The relative amount of receptor remaining on the cell surface, where MFI is mean fluorescence intensity, $[100(\text{MFI}_{\text{CCR2-T2}} - \text{MFI}_{\text{isoT2}})/(\text{MFI}_{\text{CCR2-T1}} - \text{MFI}_{\text{isoT1}})]$ at each time point was determined using a BD LSRFortessa and analyzed using FlowJo software.

cAMP assay

HEK 293 cells stably expressing hCCR2 were plated at a density of 30,000 cells per well in a 96-well tissue culture-treated plate. After an 18 h incubation at 37°C, the cells were stimulated with 100 μ M forskolin, 10 μ M MK-0812, or 1 μ M hCCL2 in the presence of 1 mM IBMX at 37°C for 15 min. Cells were then lysed, and cAMP was measured with the cAMP-Screen Direct Immunoassay System (Life Technologies, CA) according to the manufacturer's instructions.

Transient transfection

HEK 293 cells were used at 70% confluence in T175 flasks. Plasmid DNA (15 μ g) from the indicated constructs was incubated with 400 μ l of Opti-MEM I Reduced-Serum Medium (31985; Life Technologies) and electroporated using a Gene Pulser (Bio-Rad Laboratories) with the voltage set at 0.25 kV. After electroporation, cells were transferred to new T175 flasks and cultured in 25 ml of DMEM supplemented with 10% (v/v) FBS (Sigma-Aldrich).

CCR2 small interfering RNA knockdown

HEK 293 cells stably expressing hCCR2 were transfected with small interfering RNA (siRNA) for CCR2 using the DharmaFECT 1 transfection

reagent (Dharmacon, Denver, CO); 0.2 μ l of 5 μ M of siRNA and 0.5 μ l of DharmaFECT transfection reagent were added to each well of a 96-well plate containing HEK 293 hCCR2 cells. ON-TARGETplus nontargeting control pool siRNA was used as the negative control (Dharmacon). Knockdown efficiency was determined by using the QuantiGene Plex Gene Expression Assay from Thermo Fisher Scientific, following the manufacturer's instructions.

Reagents

The following small molecules were synthesized by the Medicinal Chemistry Department, of ChemoCentryx (Mountain View, CA): CCX140, CCX872, CCX598, CCX507, CCX9588, MK-0812, INCB3344, and PF04634817. Cycloheximide, IBMX, and forskolin were obtained from Sigma-Aldrich (St. Louis, MO). PTX was purchased from List Biological Lab, (Campbell, CA). Mouse and hCCL2 were both purchased from R&D Systems. YM-254890 was purchased from Wako Chemicals. Anti-hCCR2 Ab (clone number K036C2) and mouse IgG2a isotype control Ab were obtained from BioLegend (San Diego, CA). Anti-hCXCR4 Ab (clone number 12G5) was obtained from R&D systems. SMARTpool siRNA for knockdown of CCR2 and nontargeting siRNA were obtained from Dharmacon.

Results

We measured the blood plasma concentration of CCL2 in 4–6-wk-old female WT C57BL/6 mice ($n = 5$), and found it to be $\sim 60 \pm 20$ pg/ml (Fig. 1A, left). This concentration was nearly 6-fold higher in gender- and age-matched CCR2-deficient mice of the same strain

(Fig. 1A, left). Dosing WT mice with the CCR2 antagonists CCX598 (14) or MK-0812 (12, 13, 18, 19) caused plasma CCL2 levels (measured at the 4-h time point) to approach those of CCR2-deficient mice, plateauing near 300 pg/ml in CCX598 and MK-0812 dose response experiments (Fig. 1A, middle). The corresponding plasma concentrations of CCX598 and MK-0812 at this same 4-h time point are shown in Fig. 1B). A time course for assessing the concentration of each CCR2 antagonist versus CCL2 concentration in the plasma indicates that the peak CCL2 level lagged 2–4 h behind the peak antagonist level for both compounds (Fig. 1C).

To investigate immune cells as a potential source of CCL2, we cultured freshly isolated human peripheral blood monocytes in serum-containing medium for 24 h (Fig. 2). We found that CCL2 did not accumulate in the medium under these conditions (Fig. 2A). However, consistent with the *in vivo* data (Fig. 1), CCL2 accumulated in the medium when monocytes were cultured in the presence of CCR2 antagonist (Fig. 2A, left). The amount of CCL2 that accumulated in the medium correlated positively with the antagonist concentration (Fig. 2A, right). The magnitude of this effect for each antagonist paralleled its potency in a CCL2-mediated assay of monocyte chemotaxis (see Supplemental Fig. 1C for a table of potency of each antagonist in migration assays). The antagonist-driven accumulation of

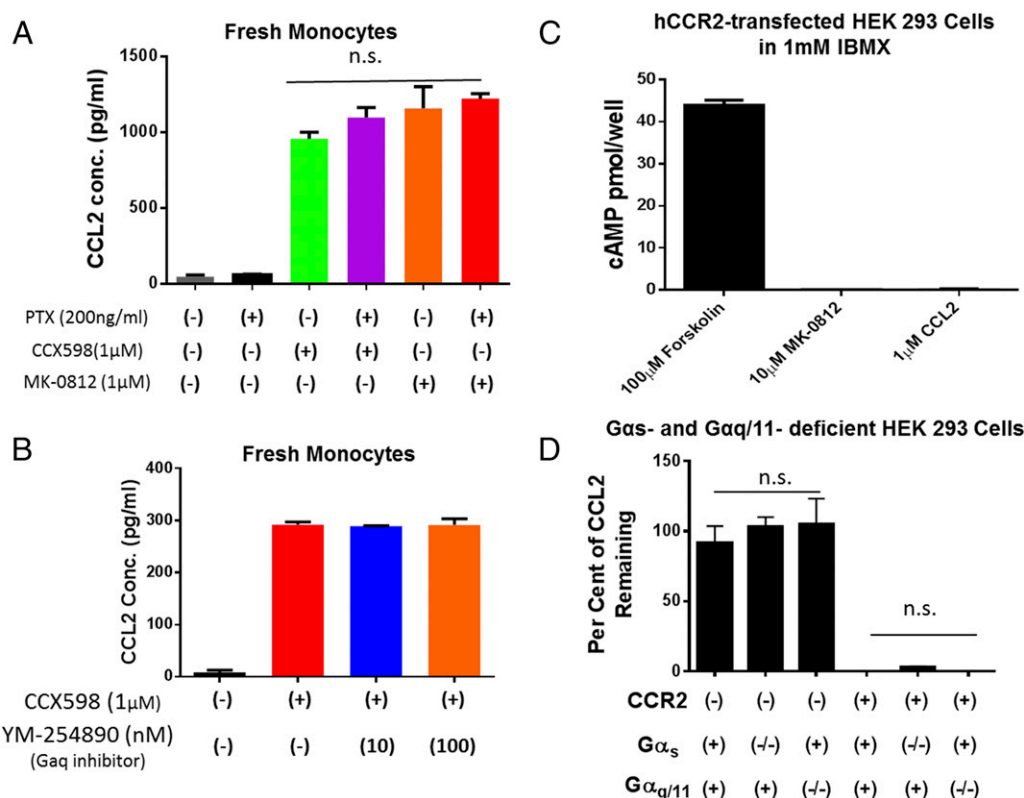


FIGURE 3. Accumulation of CCL2 following CCR2 antagonist treatment is independent of $G\alpha_i$, $G\alpha_s$, or $G\alpha_{q/11}$. **(A)** Fresh human peripheral blood monocytes were incubated for 2 h with 200 ng/ml PTX or vehicle prior to incubation with CCX598 (1 μ M), MK-0812 (1 μ M), or vehicle (DMSO), as indicated on the x-axis. PTX treatment of monocytes was demonstrated to be effective by its inhibition of their migration in a gradient of CCR2 ligands (see Supplemental Fig. 1) **(B)** Monocytes were treated with vehicle (DMSO) or $G\alpha_q$ inhibitor YM-254890 (10 or 100 nM) for 24 h in the presence of 1 μ M CCX598. **(C)** CCR2-expressing HEK 293 cells, which, like monocytes, secrete CCL2 into the medium in response to CCR2 antagonists (see Fig. 4), were stimulated with 100 μ M forskolin, CCR2 ligand CCL2 (1 μ M), or MK-0812 (10 μ M) in the presence of IBMX (1 mM). The intracellular cAMP was determined by using cAMP-Screen Direct Immunoassay System kit from (Life Technologies) **(D)** $G\alpha_s$ - and $G\alpha_{q/11}$ -deficient, CCR2- or vector-transfected HEK 293 cells were incubated with 10 μ M CCR2 antagonist or vehicle at 37°C for 24 h. The percentage of CCL2 remaining in the medium in response to CCR2 antagonist is shown as the ratio of CCL2 concentration in the CCR2 antagonist-treated medium versus vehicle (DMSO)-treated medium on the y-axis. The difference between KO $G\alpha_s$ or $G\alpha_{q/11}$ -deficient and WT cells was shown to be NS by *t* test, using GraphPad Prism (GraphPad Software). The origins of the $G\alpha_{q/11}$ - and $G\alpha_s$ -deficient cell lines are described in Refs. (15, 16), respectively. The CCL2 concentration in the conditioned medium was assessed by CCL2 ELISA from R&D Systems.

CCL2 was temperature dependent, occurring in 37°C cultures but not in 4°C cultures (Fig. 2B). Time course experiments revealed that the majority of CCL2 accumulated between 4 and 24 h (Fig. 2C). CCL2 accumulation in the medium was dependent upon new protein synthesis by the monocytes, as it did not occur in the presence of the protein synthesis inhibitor cycloheximide (Fig. 2D).

CCR2 antagonist-induced accumulation of CCL2 was not dependent on signaling through G_{α_i} , as PTX treatment of monocytes did not affect CCL2 concentration in the culture medium (Fig. 3A) despite its effectiveness in preventing chemotaxis of monocytes to CCL2 (see Supplemental Fig. 2). The G_{α_q} inhibitor YM-254890 (20, 21) had no appreciable effect on CCX598-mediated increases in CCL2 concentration (Fig. 3B). CCR2 antagonists did not cause cAMP to accumulate, suggesting that CCR2 antagonists did not induce signaling via G_{α_s} (Fig. 3C). Consistent with these results, we found that $G_{q/11}$ KO or G_{α_s} (G_s KO) (15, 16) secreted CCL2 into the medium at levels comparable to WT cells, but CCL2 was not detected in these KO lines if stably expressing CCR2 (Fig. 3D).

Although monocytes cultured with CCR2 antagonists did not secrete CCL7, CCL8, or CCL13 into the culture medium (Fig. 4A), monocytes were capable of clearing these CCR2 ligands from the medium if the ligands were added exogenously (Fig. 4B–D). This removal of exogenously added CCR2 ligands from the medium

was inhibited by CCR2 antagonists in a dose-dependent manner (Fig. 4B–D), except in the case of CCL7 (Fig. 4B). Unlike CCL8 and CCL13, CCL7 is a ligand of both CCR2 and CCR1, which are both expressed by monocytes. As such, CCR2 antagonism alone was not able to prevent CCL7 clearance from the medium. However, the addition of a CCR1 antagonist, CCX721 (17), in combination with CCX598 blocked the clearance of CCL7 (Fig. 4B), consistent with a prior report that CCR1 also removes its chemokine ligands from the medium (22).

Secretion of CCL2 into culture medium was not restricted to monocytes or immune cells. For example, the transformed human embryonic kidney cell line HEK 293 (23) constitutively secreted CCL2 but not the other known CCR2 ligands, CCL7, CCL8, or CCL13 (Fig. 5A). When HEK 293 cells were transfected with hCCR2, accumulation of CCL2 was greatly reduced in the medium (Fig. 5B). However, addition of CCR2 antagonists to CCR2-expressing HEK 293 cells caused CCL2 to accumulate in a dose-dependent manner, similar to that demonstrated for monocytes in Fig. 2 (Fig. 5B). To eliminate potential artifacts implicit in comparing a single transfected HEK 293 clone to a single WT clone, we created transient CCR2 and CCR5 transfectants (Fig. 5C). CCL2 did not accumulate in the medium of transient CCR2 transfectants unless a CCR2 antagonist was present (Fig. 5C). CCR5 transfectants secreted CCL2 independently of whether a CCL2 antagonist was present (Fig. 5C, right

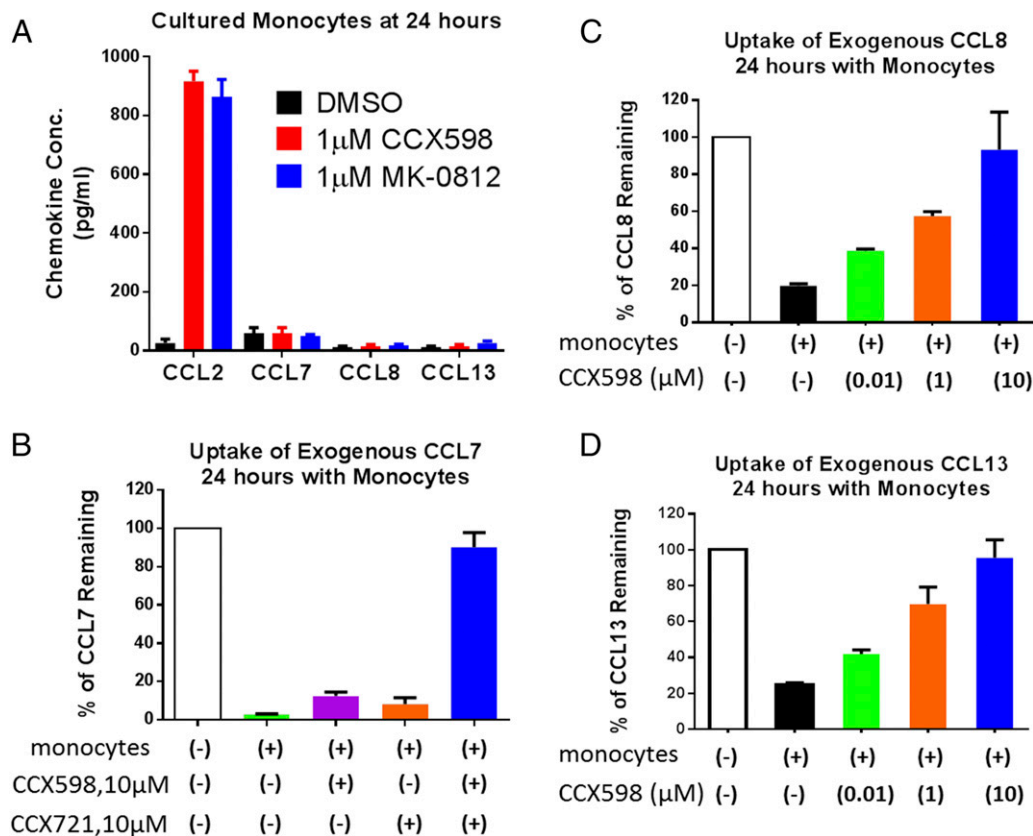


FIGURE 4. Antagonism of CCR2 on monocytes does not induce accumulation of CCR2 ligands other than CCL2, but monocytes are able to deplete exogenously added CCL7, CCL8, and CCL13 from culture medium. **(A)** Fresh human peripheral blood monocytes were incubated with vehicle (DMSO, black bars), CCX598 (1 μM) (red bars), or MK-0812 (1 μM) (blue bars). Cell culture supernatant was collected at 24 h, and the level of CCL2, CCL7, CCL8, and CCL13 were measured by ELISA. Data from a single experiment with cells from a single donor are shown, representative of at least two other donors. **(B)** 1 nM MCP-3/CCL7 was added to the culture medium with or without CCX598 (10 μM) and/or CCX721 (a CCR1 antagonist) in the presence (black bars) or absence (white bar) of monocytes. **(C and D)** A total of 1 nM MCP-2/CCL8 or MCP-4/CCL13 with the indicated concentrations of CCX598 were added to the culture medium in the presence (black bars) or absence (white bars) of monocytes. Medium was collected after 24 h of culture, and the remaining CCL7, CCL8, and CCL13 were measured by ELISA. The percentage of remaining exogenously added ligand was normalized to the ligand concentration measured from monocyte-free control medium (white bars).

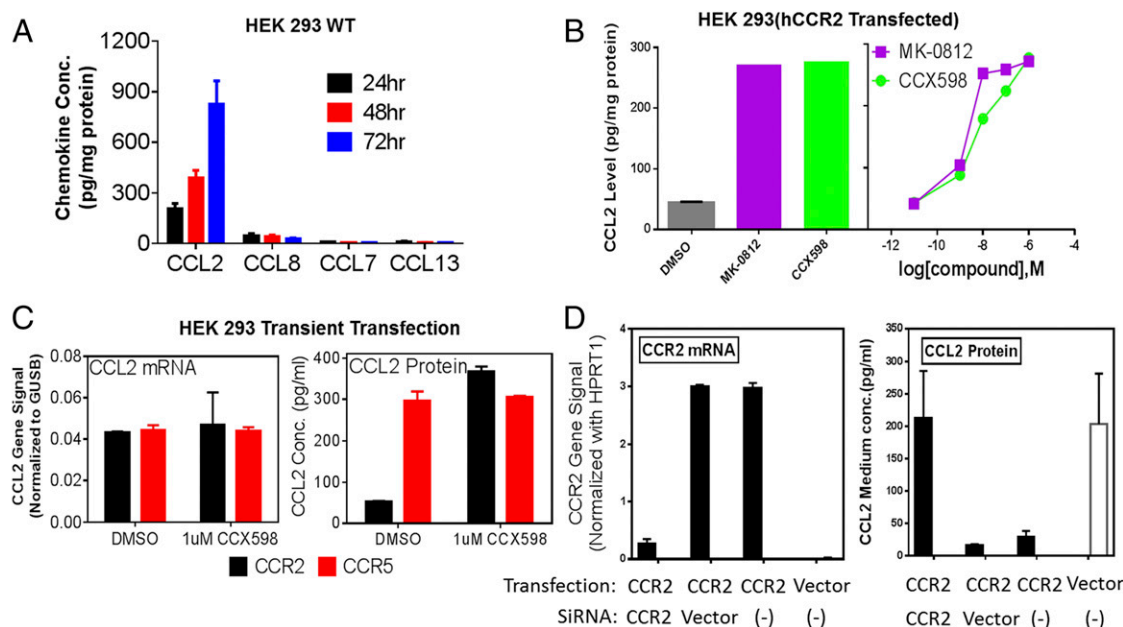


FIGURE 5. CCL2 is constitutively expressed by WT HEK 293 cells but is cleared from the medium by hCCR2-transfected HEK 293 cells. **(A)** WT HEK 293 cells were cultured in DMEM supplemented with 10% FBS for 24 h (black bars), 48 h (red bars), or 72 h (blue bars) hours. The concentration of each of the four known CCR2 ligands in the culture medium (CCL2, CCL7, CCL8, and CCL13) was determined by ELISA. **(B)** Left panel, The same experiment as in (A) (at 24 h) was performed with hCCR2-expressing HEK 293 cells instead of WT HEK 293 cells, and it was performed in the presence of DMSO or CCR2 antagonists MK-0812 (purple squares) and CCX598 (green circles) on CCL2 concentrations in medium from hCCR2-expressing HEK 293 cells. Data shown in (B) are from a single experiment representative of three additional experiments. **(C)** WT HEK 293 cells were transiently transfected with expression vectors for either CCR2 (black bars) or CCR5 (red bars). Twenty-four hours after transfection, each cell type was plated (at identical density) in 24-well plates and treated with CCX598 (1 μ M) or vehicle control (DMSO) for another 24 h. Forty-eight hours posttransfection, the cultured media were collected (to measure CCL2 protein concentration), and cells were collected (to measure CCL2 mRNA expression). Cells were lysed, and CCL2 mRNA was measured with the QuantiGene Plex Gene Expression Assay. **(D)** Stably transfected hCCR2-expressing HEK 293 cells (or vector transfectants) were transiently transfected with the indicated siRNA and incubated for 48 h. mRNA expression levels for CCR2 were determined by the QuantiGene Plex Gene Expression Assay and normalized to HPRT1 mRNA levels. The CCL2 concentrations in the culture medium 48 h after siRNA knock down were determined by ELISA.

panel, red bars). Both CCR2 and CCR5 transient transfectants expressed CCL2 mRNA at equal levels whether CCR2 antagonist was present (Fig. 5C, left panel). We further assessed the direct role of transfected CCR2 in preventing CCL2 accumulation by inhibiting CCR2 mRNA expression via siRNA in HEK 293 cells (Fig. 5D, left panel). Consistent with earlier results, CCL2 accumulated in the medium of siRNA-treated CCR2-transfected cultures but not in the absence of siRNA treatment or with empty siRNA vector treatment (Fig. 5D, right panel).

A clue toward the mechanism of chemokine clearance by CCR2 is provided by certain atypical chemokine receptors (ACKR). By constitutive internalization, these receptors transport the chemokine into the cell for degradation and recycle back to the surface for further rounds of chemokine depletion (recently reviewed in Ref. 24). We therefore investigated the trafficking behavior of CCR2 and its kinetics. Using a postlabel protocol for detecting surface levels of CCR2, in which Ab is added at various time points after treatment, we observed that CCR2 levels on the monocyte surface remained constant whether the samples were incubated at 4°C (internalization is inhibited) or 37°C (permissive to internalization). Treatment with CCR2 antagonists or PTX showed no difference from control with medium alone (Fig. 6A, left panel). By contrast, using a prelabel protocol in which Ab labeling is done once at the start of the experiment, we observed only <5% of the labeled receptor remaining on the cell surface following 30 min incubation at 37°C (Fig. 6A, right panel). The presence of CCR2 antagonist did not affect this internalization nor did the inactivation of G α_i via PTX treatment (Fig. 6A, right panel). A time course showed

that constitutive internalization of CCR2 reached its maximum at 30 min (Fig. 6B). Together, these data suggest that CCR2 undergoes constitutive internalization (Fig. 6A, right panel) concurrent with constitutive recycling or replenishment of the surface receptor from intracellular stores (Fig. 6A, left panel). This ensures a constant level of cell surface receptors and provides a robust mechanism for chemokine depletion from the medium. Although antagonists have no effect on constitutive internalization and recycling of CCR2, inhibition of chemokine binding prevents uptake and leads to accumulation of CCL2 in the extracellular space.

We next investigated whether constitutive internalization is a general characteristic of chemokine receptors on monocytes or restricted to a subset such as CCR2 and CCR1 (22) (Fig. 7). CXCR4 is a chemokine receptor expressed along with CCR2 on the cell surface of monocytes, and its only known ligand, CXCL12, induces monocyte migration (25). CXCL12 did not accumulate in the culture medium of fresh human monocytes in the presence or in the absence of the CXCR4 antagonist AMD 3100 (Fig. 7A). We found that, unlike CCR2 ligands, monocytes did not remove appreciable amounts of the CXCR4 ligand from the culture medium, even in the absence of antagonist AMD 3100 (Fig. 7B). We took advantage of the fact that saturated staining of human monocytes with anti-CXCR4 mAb yields an MFI similar to that of anti-CCR2 (Fig. 7C), allowing us to directly compare internalization of CCR2 and CXCR4 ligands within a given population of cells. By prelabeling cell surface CXCR4 with fluorescent mAb prior to incubation at 37°C (as done for CCR2 in Fig. 6), we observed that monocyte

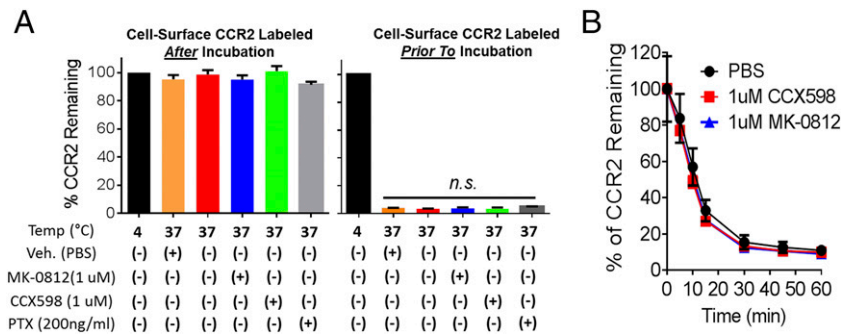


FIGURE 6. CCR2 is constitutively internalized on human monocytes, and this process is not affected by PTX or by CCR2 antagonists. **(A)** Left panel, Fresh human peripheral blood monocytes were incubated for 30 min at 37°C with CCR2 antagonists, PBS control, or cells were pretreated with PTX (as indicated in the matrix on the x-axis), then labeled for flow cytometry with fluorescently labeled anti-CCR2 mAb (postlabel protocol). Percentage of specific CCR2 MFI on the cell surface with respect to that of cells kept at 4°C is shown on the y-axis. Right panel, Experiment identical to that shown in the left panel, except cells were labeled with the anti-CCR2 mAb before 30 min incubation at 37°C (prelabel protocol). The variance between the untreated control, PBS, CCX598, MK-0812, and PTX-treated cells was shown to be NS by ANOVA with the *t* test, using GraphPad Prism (GraphPad Software). The data were generated from three replicates shown as mean $\pm$ SEM. **(B)** Cells were labeled with anti-CCR2 mAb on ice as in the right panel of (A) and pretreated with or without CCX598 (1 μ M) or MK-0812 (1 μ M) for 1 h, and then incubated at 37°C for the amount of time shown on the x-axis. The amount of receptor-specific MFI remaining on the cell surface is shown on the y-axis.

cell surface CXCR4 does not undergo constitutive internalization (Fig. 7C, 7D).

Discussion

Human clinical trials and preclinical models demonstrate in multiple species that in vivo treatment with CCR2 antagonists leads to a significant, reproducible increase in plasma CCL2 (3, 6, 10, 11). Several putative explanations for this observation have been proposed, including the following: CCR2 antagonists simply displace

existing CCL2 already bound to CCR2 (6); antagonists block clearance of CCL2 from the blood via ACKR1 (3, 10); CCR2 antagonists induce increased production of CCL2 by activating alternative signaling pathways from CCR2 (6, 10); or that CCR2 itself controls homeostasis of CCL2 (11). In terms of alternative signaling pathways, we ruled out the dependence of antagonist-induced accumulation of CCL2 on $G\alpha_i$, the main G protein that couples to CCR2, as well as $G\alpha_s$ and $G\alpha_{q/11}$. The finding that CCR2^{-/-} mice inefficiently clear excess CCL2 from the blood

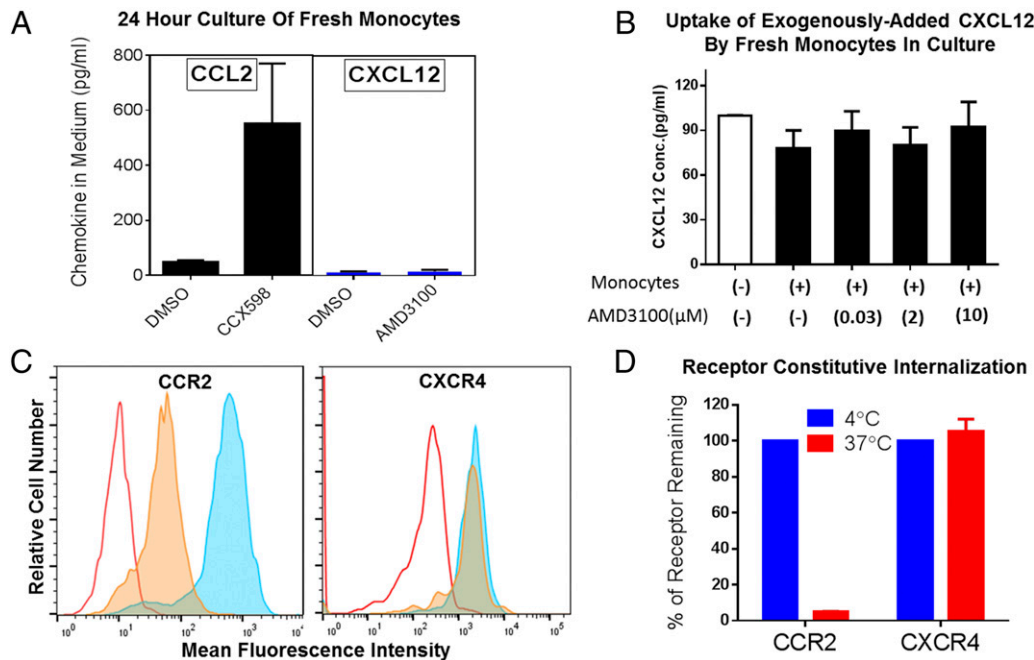


FIGURE 7. Constitutive internalization of cell surface receptors and constitutive uptake of ligand are not universal characteristics of chemokine receptors. **(A)** Freshly isolated monocytes were cultured in RPMI 1640 supplemented with 10% FBS in the presence or absence of CCR2 antagonist (CCX598, 1 μ M, black bar), CXCR4 antagonist (AMD3100, 1 μ M, blue bar), or DMSO control for 24 h. The CCL2 and CXCL12 concentration in the culture supernatant were measured by ELISA (R&D Systems). **(B)** CXCL12 (1 nM) and the indicated concentrations of AMD3100 were added to the culture medium in the presence (black bars) or absence (white bar) of freshly isolated monocytes. Media were collected after 24 h of culture, and the concentration of CXCL12 remaining in the medium was measured by ELISA. The percentage of exogenously added CXCL12 was normalized to the ligand concentration measured from cell-free control medium. **(C)** Constitutive internalization of cell surface CCR2 and CXCR4 on monocytes was measured as described in Fig. 6. Representative flow cytometry plots showing initial cell surface CXCR4 or CCR2 (solid blue histograms) remaining after 30 min of ligand-independent internalization at 37°C (solid orange histograms) and isotype-matched control (open red histogram). **(D)** Quantitation of receptor-specific MFI from experiment in (C), performed in triplicate.

at rates comparable to those of antagonist-treated WT mice (Fig. 1) (26) renders the possibility of ACKR1 involvement improbable, as ACKR1 expression is unlikely to be affected by CCR2 gene deletion [their genes are located on different chromosomes (27)]. Finally, the lag time between antagonist dosing and peak CCL2 plasma levels (2–4 h, Fig. 1C) is not consistent with simple displacement of CCL2 from its receptor.

In the current study, we discovered that this phenomenon can be recapitulated in vitro in cultures of freshly isolated human monocytes. CCL2 accumulates in the culture medium in a dose-dependent manner in the presence of CCR2 antagonists but not in the absence of such antagonists. The same is true for the HEK 293 cell line, which constitutively secretes CCL2 into the culture medium. However, when HEK 293 cells are stably or transiently transfected to express CCR2, CCL2 ceases to accumulate. Thus, the combined effects of constitutive CCL2 secretion by monocytes and other cell types and constitutive CCL2 uptake by CCR2 can parsimoniously explain both in vivo and in vitro observations. Consistent with this uptake, we found that CCR2 constitutively internalizes and recycles and that its turnover is dynamic and rapid, with the entire cohort of CCR2 molecules on the surface of a given monocyte being replaced every 30 min. Although constitutive internalization of CCR2 is not affected by the presence of CCR2 antagonists, these antagonists do prevent CCR2 from transporting CCL2 into the cell.

The notion of homeostatic chemokine internalization and clearance from the blood and microenvironments has been a familiar concept in the realm of ACKRs for many years and is thought to constitute a major role for such receptors in immunology and cell trafficking (24). Interestingly constitutive internalization and chemokine clearance has also been observed for some G protein-coupled chemokine receptors such as CCR1 (22). CCR2 has also been reported to take up chemokine from the medium (28). Additionally, because KO mice of CXCR3 and CXCR2 show elevated serum levels of their respective ligand(s), these receptors may have a similar propensity to clear their ligands from the extracellular space (29). By contrast, we found that CXCR4 is not constitutively internalized, and does not mediate the clearance of its ligand, CXCL12 (Fig. 7). Thus, it is unclear why some chemokine receptors show this behavior, whereas others do not, but we speculate that it has to do with the need for tight regulation of the extracellular concentrations of their cognate ligands. If true, this would indicate that the body places a high premium on maintaining very low blood levels of CCL2, perhaps so that monocytes and other CCR2-expressing cells can sense subtle CCL2 gradients, and migrate over long distances to sites of inflammation or injury without CCR2 desensitization. This also suggests that CCR2 effectively acts as a dual-function receptor that is capable of promoting cell migration through G protein-mediated signaling pathways as well as chemokine clearance through G protein-independent constitutive internalization and recycling.

In conclusion, we have demonstrated that the most likely explanation for the phenomenon of antagonist-induced increases of CCL2 in vivo in the circulation is the constitutive secretion of CCL2 by monocytes and other cell types, whose uptake by CCR2 is blocked by the administered antagonist, in a manner that does not require receptor signaling. Because the other CCR2 ligands are not constitutively secreted, the overall mechanism may be specific to the CCL2/CCR2 axis, which may have clinical implications.

Disclosures

B.N.Z., J.J.C., L.S.E., Y.W., S.Y., T.D., Y.Z., J.P.M., A.K., P.Z., I.F.C., R.S., and T.J.S. are full-time employees and shareholders of ChemoCentryx, Inc., which funded this research. The other authors have no financial conflicts of interest.

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